Topic C3: Chemical reactions

C3.1 Introducing chemical reactions

Summary
A chemical equation represents, in symbolic terms, the overall change in a chemical reaction. New materials are formed through chemical reactions but mass will be conserved. This can be explained by a model involving the rearrangement of atoms. Avogadro gave us a system of measuring the amount of a substance in moles.

Underlying knowledge and understanding
Learners should be familiar with chemical symbols and formulae for elements and compounds. They should also be familiar with representing chemical reactions using formulae and equations. Learners will have knowledge of conservation of mass, changes of state and chemical reactions.

Common misconceptions
Although learners may have met the conservation of mass they still tend to refer to chemical reactions as losing mass. They understand that mass is conserved bu not the number or species of atoms. They may think that the original substance vanishes 'completely and forever' in a chemical reaction.

Tiering
Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
CM3.1i	arithmetic computation and ratio when determining empirical formulae, balancing equations	M1a, M1c
CM3.1ii	**calculations with numbers written in standard form when using the Avogadro constant**	**M1b**
CM3.1iii	provide answers to an appropriate number of significant figures	M2a
CM3.1iv	convert units where appropriate **particularly from mass to moles**	M1c

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	**To include**	**Maths**	**Working scientifically**	
C3.1a use chemical symbols to write the formulae of elements and simple covalent and ionic compounds		M1a, M1c	WS1.4a	
C3.1b use the names and symbols of common elements and compounds and the principle of conservation of mass to write formulae and balanced chemical equations **and half equations**		M1a, M1c	WS1.4c	
C3.1c use the names and symbols of common elements from a supplied Periodic Table to write formulae and balanced chemical equations where appropriate	the first 20 elements, Groups 1, 7, and 0 and other common elements included within the specification			
C3.1d use the formula of common ions to deduce the formula of a compound		M1a, M1c		
C3.1e construct balanced ionic equations		M1a, M1c		
C3.1f describe the physical states of products and reactants using state symbols (s, l, g and aq)				
C3.1g recall and use the definitions of the Avogadro constant (in standard form) and of the mole	**the calculation of the mass of one atom/molecule** In recognition of IUPAC's review, we will accept both the classical (carbon-12 based) and revised (Avogadro constant based) definitions of the mole in examinations from June 2018 onwards (see https://iupac.org/new-definition-mole-arrived/)	M1b, M1c	WS1.4b, WS1.4c, WS1.4d, WS1.4f	
C3.1h explain how the mass of a given substance is related to the amount of that substance in moles and vice versa		M1c, M2a	WS1.4b, WS1.4c	

C3.1i	recall and use the law of conservation of mass			WS1.4c	
C3.1j	explain any observed changes in mass in non-enclosed systems during a chemical reaction and explain them using the particle model			WS1.1b, WS1.4c	
C3.1k	**deduce the stoichiometry of an equation from the masses of reactants and products and explain the effect of a limiting quantity of a reactant**		M1c	WS1.3c, WS1.4c, WS1.4d, WS1.4f	
C3.1l	**use a balanced equation to calculate masses of reactants or products**		M1c	WS1.3c, WS1.4c	